



Smart Pharmacy Inventory Optimization

End-to-end analytics system reducing stock-outs and expiry losses in healthcare pharmacy operations

Business Goal



Reduce Stock-Outs

Minimize medicine shortages and revenue leakage



Minimize Losses

Reduce expiry-related financial waste



Improve Turnover

Optimize inventory velocity and efficiency



Real-Time Visibility

Enable data-driven replenishment planning



Problem Statement

Challenges Faced

- Frequent medicine shortages
- Excess inventory accumulation
- High expiry losses
- Reactive manual monitoring
- Poor demand visibility

Business Impact

Unplanned stock-outs of critical medicines caused revenue leakage from unavailable drugs and inefficient reorder planning.

Technology Stack



Python Analytics

Pandas, NumPy, SciPy, SQLAlchemy for data processing and inventory intelligence



SQL Database

MySQL for centralized storage of 250K+ inventory records



Power BI

Interactive dashboards for executive monitoring and operational decisions



ETL Automation

GitHub Actions for scheduled API extraction and data refresh pipelines

Methodology



Data Ingestion

API extraction from healthcare ERP systems with automated refresh pipelines



Data Cleaning

Validation, deduplication, and standardization of drug names and classes



Exploratory Analysis

Trend analysis of stock movement, seasonal demand, and expiry patterns



Inventory Intelligence

ABC classification, reorder point calculation, and safety stock recommendations



Financial Modeling

Revenue loss quantification from stock-outs and expired inventory



Dashboard Development

Interactive Power BI dashboards for real-time operational monitoring

Core Deliverables

1

Revenue Loss Detection

Financial models measuring losses from expired medicines and stock-out revenue leakage

2

Seasonal Demand Intelligence

Historical sales analysis detecting seasonal spikes and forecasting demand trends

3

Smart Reorder Engine

Dynamic logic recommending reorder timing, quantities, and safety stock levels

4

Inventory Risk Heatmap

Visual tracking of stock levels with color-coded risk indicators for fast-moving medicines

5

ABC Inventory Analysis

Medicine segmentation into high-value, moderate-demand, and low-value categories



Key Insights

Operational Insights

- Antibiotics and vaccines showed highest stock-out frequency
- Seasonal demand spikes for respiratory medicines
- Critical medicines understocked despite consistent demand
- Slow-moving specialties caused expiry losses

Financial Insights

- Expiry losses concentrated in high-cost oncology drugs
- Stock-outs created hidden revenue leakage
- Small percentage generated majority of revenue
- Overstocking increased storage costs

Recommendations



Inventory Optimization

Higher safety stock for critical medicines, automated reorder alerts, reduced low-demand procurement



Expiry Reduction

FEFO rotation, expiry monitoring dashboards, optimized procurement cycles



Demand Planning

Predictive replenishment using consumption trends and seasonal disease patterns



Operational Automation

Automated refresh pipelines, centralized reporting, real-time inventory monitoring



Business Impact

Operational Impact

- Improved inventory visibility
- Faster stock shortage identification
- Better replenishment planning
- Reduced manual monitoring

Financial Impact

- Reduced expiry losses
- Improved revenue retention
- Lower emergency procurement
- Optimized carrying costs

Strategic Impact

- Data-driven pharmacy management
- Improved medicine availability
- Scalable analytics framework

Skills Demonstrated

- Healthcare Analytics & Business Intelligence
- Inventory Optimization & Forecasting
- Financial Loss Analysis
- ETL Pipeline Development & API Integration
- SQL Database Management
- Power BI Dashboarding & Data Visualization
- Python Data Analytics
- Process Automation